

Symmetrizing the defect gives nonlinear real interpolation

Full-result packet for arXiv:2212.01262

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Status. Likely-valid full positive answer to the stated interpolation question, pending human review. The abstract interpolation theorem and its K -functional proof are complete. Novelty confidence is low because sublinear interpolation itself is classical; the key new identification is the symmetrized approximation defect.

1 Source question

Gal and Niculescu ask in Section 6, p. 11, of [1]:

Extend the classical interpolation technique from linear continuous operators to sublinear, monotone and strongly translatable operators.

They explain that a positive answer should permit the Berens–DeVore interpolation method [2] to replace the weaker interior- L^p estimate obtained by their direct Swetits–Wood argument.

$k=0$

and

$$|T_n(f)(x) - f(x)| \leq |T_n(f)(x) - B_n(f)(x)| + |B_n(f)(x) - f(x)|.$$

By the result in Swetits–Wood applied to the positive linear operators $B_n(f)(x)$, for $1 < p < \infty$, it immediately follows

$$\|T_n(f) - f\|_p \leq M_p \left(\frac{1}{n} \|f\|_p + \omega_{2,p}(f; 1/\sqrt{n})_p \right) + \omega(f; 1/(n+1)),$$

which clearly it is essentially better than the estimate in the previous section.

These considerations suggest that possibly the shortcoming in Remark 5 is due to the method of proof in Swetits–Wood [18] and raise the following.

Open Question. Extend the classical interpolation technique from linear continuous operators to sublinear, monotone and strongly translatable operators.

A positive answer would allow to essentially improve the estimate in Theorem 4 by following now the method of proof in Theorem 3 of Berens–DeVore [7].

Figure 1: The open question and its intended quantitative application in the source paper [1, p. 11].

2 Setting and proof intuition

A map T between vector lattices is *order-sublinear* if

$$T(f + g) \leq T(f) + T(g), \quad T(\alpha f) = \alpha T(f) \quad (\alpha \geq 0).$$

This is the notion called sublinearity in [1]. It is one-sided, so the approximation defect $Tf - f$ is not immediately a standard sublinear operator after taking absolute values.

Proof intuition. Let A be the linear operator being approximated and put $Df = Tf - Af$. Order-sublinearity gives $0 = D0 \leq Df + D(-f)$, hence the negative part of Df is controlled by $D(-f)$. Therefore

$$Qf = Df \vee D(-f)$$

is nonnegative and dominates $|Df|$. It is also even and subadditive, so it is exactly the conventional sublinear envelope needed for real interpolation. The K -functional estimate then follows from one lattice decomposition: if $0 \leq h \leq a + b$, split $h = (h \wedge a) + (h - a)^+$.

3 The interpolation theorem

Let (X_0, X_1) and (Y_0, Y_1) be compatible couples of Banach lattices. Their real-interpolation spaces are denoted by $(X_0, X_1)_{\theta, q}$ and $(Y_0, Y_1)_{\theta, q}$, with the usual K -method normalization [3]. We use only the ideal property of Banach lattices: $0 \leq u \leq v \in Y_i$ implies $u \in Y_i$ and $\|u\|_{Y_i} \leq \|v\|_{Y_i}$.

Theorem 1 (Symmetrized-defect interpolation). *Let $T : X_0 + X_1 \rightarrow Y_0 + Y_1$ be order-sublinear and let $A : X_0 + X_1 \rightarrow Y_0 + Y_1$ be linear. Set*

$$Df = Tf - Af.$$

Assume that for $i = 0, 1$,

$$\|Df\|_{Y_i} \leq a_i \|f\|_{X_i} \quad (f \in X_i), \quad (1)$$

where $a_i > 0$. Then, for $0 < \theta < 1$ and $1 \leq q \leq \infty$,

$$\|Tf - Af\|_{(Y_0, Y_1)_{\theta, q}} \leq 2a_0^{1-\theta} a_1^\theta \|f\|_{(X_0, X_1)_{\theta, q}}. \quad (2)$$

Thus the usual real-interpolation technique extends to the source paper's class. Neither monotonicity nor strong translatability is needed for the interpolation step itself.

Proof. The defect $D = T - A$ is order-sublinear because A is linear. Define

$$Qf = Df \vee D(-f).$$

Since $0 = D0 \leq Df + D(-f)$, we have $-Df \leq D(-f)$ and consequently

$$0 \leq |Df| \leq Qf. \quad (3)$$

Also $Q(-f) = Qf$. For $\alpha \geq 0$, $Q(\alpha f) = \alpha Qf$, and evenness gives absolute homogeneity for all real scalars. Finally,

$$D(f + g) \leq Df + Dg \leq Qf + Qg$$

and the same inequality applied to $-(f + g)$ shows that

$$Q(f + g) \leq Qf + Qg. \quad (4)$$

Thus Q is a conventional nonnegative sublinear operator.

Moreover,

$$Qf \leq |Df| + |D(-f)|,$$

so (1) implies

$$\|Qf\|_{Y_i} \leq 2a_i \|f\|_{X_i}. \quad (5)$$

We now give the K -functional step explicitly. Fix a decomposition $f = f_0 + f_1$ with $f_i \in X_i$, and write $h = Qf$, $b_i = Qf_i$. By (4), $0 \leq h \leq b_0 + b_1$. The Riesz decomposition

$$h_0 = h \wedge b_0, \quad h_1 = h - h_0 = (h - b_0)^+$$

satisfies $h = h_0 + h_1$, $0 \leq h_0 \leq b_0$, and $0 \leq h_1 \leq b_1$. Hence, by the ideal property and (5),

$$\begin{aligned} K(t, Qf; Y_0, Y_1) &\leq \|h_0\|_{Y_0} + t\|h_1\|_{Y_1} \\ &\leq 2a_0\|f_0\|_{X_0} + 2ta_1\|f_1\|_{X_1}. \end{aligned}$$

Taking the infimum over all decompositions gives

$$K(t, Qf; Y_0, Y_1) \leq 2a_0 K\left(t \frac{a_1}{a_0}, f; X_0, X_1\right). \quad (6)$$

Multiplying by $t^{-\theta}$, using the change of variable $u = ta_1/a_0$, and taking the $L^q(du/u)$ norm yields

$$\|Qf\|_{(Y_0, Y_1)_{\theta, q}} \leq 2a_0^{1-\theta} a_1^\theta \|f\|_{(X_0, X_1)_{\theta, q}}.$$

Equation (2) follows from (3) and the ideal property of the interpolated Banach lattice. If one endpoint constant is zero, the same conclusion follows by replacing it by a positive ε and letting $\varepsilon \downarrow 0$. $\square$

Corollary 2 (Strong-type interpolation of approximation errors). *Suppose $T - A$ acts consistently on $L^{p_0} + L^{p_1}$ and*

$$\|Tf - Af\|_{p_i} \leq a_i \|f\|_{p_i} \quad (i = 0, 1),$$

where $1 \leq p_0 < p_1 \leq \infty$. If $1/p = (1 - \theta)/p_0 + \theta/p_1$, then

$$\|Tf - Af\|_p \leq C_{p_0, p, p_1} a_0^{1-\theta} a_1^\theta \|f\|_p.$$

In particular, endpoint approximation rates interpolate geometrically even though T is only order-sublinear.

Proof. Apply Theorem 1 to the usual real-interpolation identities for Lebesgue spaces. The harmless factor 2 is absorbed into the displayed constant. $\square$

4 Quantitative Korovkin consequence

We now insert the endpoint estimates already proved in [1]. Let $e_j(x) = x^j$ on $[0, 1]$ and define, for $r = 1, \infty$,

$$\lambda_r(T) = \max\{\|T(e_0) - e_0\|_r, \|T(e_1) - e_1\|_r, \|T(-e_1) + e_1\|_r, \|T(e_2) - e_2\|_r\}. \quad (7)$$

Corollary 3 (Interior- L^p Korovkin interpolation). *Let T be one consistent monotone, order-sublinear, strongly translatable operator on the endpoint spaces used below, satisfying the hypotheses of [1, Theorem 2]. For $1 < p < \infty$, put*

$$\theta = 1 - \frac{1}{p}, \quad Z_p = (W_1^3([0, 1]), W_\infty^2([0, 1]))_{\theta, p}, \quad \eta_p = \lambda_1(T)^{1/p} \lambda_\infty(T)^{1-1/p}.$$

Then

$$\|Tf - f\|_p \leq C_p \eta_p \|f\|_{Z_p} \quad (f \in Z_p). \quad (8)$$

The same bound holds with $\|Tf - f\|_p$ replaced by $\|Qf\|_p$.

If, additionally, $T : L^p \rightarrow L^p$ is bounded with norm at most R_p , then for every $f \in L^p$,

$$\|Tf - f\|_p \leq (R_p + 1)K\left(\frac{C_p \eta_p}{R_p + 1}, f; L^p, Z_p\right). \quad (9)$$

Consequently, a uniformly L^p -bounded sequence (T_n) with $\lambda_1(T_n), \lambda_\infty(T_n) \rightarrow 0$ converges to the identity on all of L^p , with the quantitative rate in (9).

Proof. The source estimate [1, Theorem 2] gives

$$\|Tf - f\|_r \leq C \lambda_r(T) \|f\|_{W_\infty^2}, \quad r = 1, \infty.$$

For the L^1 endpoint use the one-dimensional Sobolev embedding $W_1^3 \hookrightarrow W_\infty^2$; for the L^∞ endpoint use W_∞^2 directly. Theorem 1, with $(Y_0, Y_1) = (L^1, L^\infty)$ and $(L^1, L^\infty)_{1-1/p, p} = L^p$, proves (8) and, by the proof of Theorem 1, the asserted bound for Qf .

For the second claim, monotonicity gives, for every $h \in L^p$,

$$Qh \leq T(|h|) + |h|, \quad \|Qh\|_p \leq (R_p + 1)\|h\|_p.$$

For $g \in Z_p$, subadditivity of Q and (8) therefore imply

$$\|Tf - f\|_p \leq \|Q(f - g)\|_p + \|Qg\|_p \leq (R_p + 1)\|f - g\|_p + C_p \eta_p \|g\|_{Z_p}.$$

Taking the infimum over g is exactly (9). Smooth functions belong to Z_p and are dense in L^p , so the final convergence statement follows. $\square$

Remark 4 (The nonlinear Bernstein example). For $T_n(f) = \max\{B_n(f), B_{n+1}(f)\}$ from [1, Example 1], the only nonzero test error is

$$T_n(e_2) - e_2 = \frac{x(1-x)}{n}.$$

Thus $\lambda_1(T_n) = 1/(6n)$ and $\lambda_\infty(T_n) = 1/(4n)$, so Corollary 3 gives

$$\|T_n f - f\|_p \leq \frac{C_p}{n} \|f\|_{Z_p}$$

and an all- L^p K -functional estimate at parameter $1/n$. This recovers, through one nonlinear interpolation principle, the sharper behavior that the source paper obtains for this example by decomposing it back into the two linear Bernstein operators.

5 Verification, limitations, and novelty

Verification. The proof is algebraic except for standard Banach-lattice and real- interpolation identities. The critical checks are: (i) the inequality $-Df \leq D(-f)$; (ii) subadditivity of the join defining Q ; (iii) the Riesz split $h = (h \wedge b_0) + (h - b_0)^+$; and (iv) the scaling $a_0^{1-\theta} a_1^\theta$ in (6). Each is displayed in the proof. The Bernstein test values were also recomputed directly from $B_n(e_2) = e_2 + x(1-x)/n$.

Scope. Theorem 1 fully answers the question of whether the classical real-interpolation mechanism extends to the stated nonlinear class. It does not claim that an arbitrary desired sharp Korovkin formula follows without work: one must still prove the endpoint bounds appropriate to that formula. Corollary 3 is the direct consequence of the endpoint bounds actually present in the source paper. Consistency of the same operator on the two endpoint spaces is essential.

Bounded novelty search. A search through 9 August 2026 used arXiv:2212.01262, its exact open-question sentence, the published 2024 title and DOI, and the phrases *symmetrized defect*, *order-sublinear interpolation*, *monotone strongly translatable interpolation*, and *nonlinear Korovkin interpolation*. No explicit answer using the map $Df \vee D(-f)$ was found. The published 2024 version [1] still states the question. General interpolation of conventional sublinear maps is classical [3], so novelty confidence is *low*: the potentially new item is the short defect-symmetrization bridge and its Korovkin specialization, not the interpolation theorem after that bridge.

Human-review recommendation. Verify that the intended applications use one operator consistently defined on both endpoints, check the endpoint-domain choice in Corollary 3, and search later citing papers for the same symmetrized-defect observation before making a novelty claim.

References

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